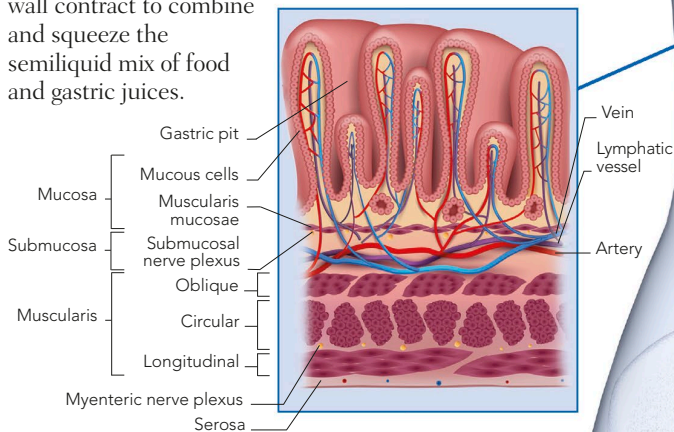


STOMACH AND SMALL INTESTINE

THE STOMACH DIGESTS FOOD CHEMICALLY AND PHYSICALLY. THE SMALL INTESTINE CONTINUES THE CHEMICAL BREAKDOWN AND ABSORBS NUTRIENTS.

STOMACH STRUCTURE

The stomach is a muscular-walled, J-shaped sac in which food is stored, churned, and mixed with gastric juices secreted by its lining. The juices include digestive enzymes and hydrochloric acid, which breaks down food and kills potentially harmful microbes. The smooth muscles of the wall contract to combine and squeeze the semiliquid mix of food and gastric juices.



Longitudinal

Circular

Oblique

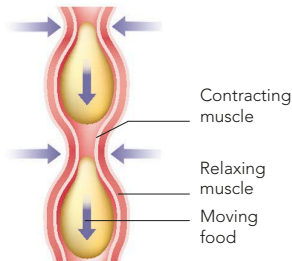
The three muscle layers cause the stomach to twist and writhe into almost any shape

LAYERS OF THE STOMACH WALL

The stomach wall has four main layers; the mucosa, submucosa, muscularis, and serosa. The mucosa has deep infolds (gastric pits) that contain the gastric glands, and cells that produce acid (parietal cells), enzymes (zymogenic or chief cells), and hormones (neuroendocrine cells).

PERISTALSIS

Waves of muscle contraction propel food through the digestive tract. The circular muscle contracts and relaxes in sequence, producing a "travelling wave" known as peristalsis.



Duodenum

First and shortest section of the small intestine, about 25cm (10in) long